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Does short-term gluteal activation enhance muscle performance?

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ABSTRACT

A reduction in gluteus maximus (GM) strength may contribute to the etiology of musculoskeletal impairments and lower-extremity injuries. Currently, there is a paucity of evidence regarding the efficacy of implementing a short-term gluteal activation programme to improving muscle performance. Twenty four semi-professional rugby males were assigned randomly to a gluteal activation group (GLUTE) or a control group (CON). During the 6-week training intervention, the GLUTE and CON groups performed the same training, however that GLUTE group performed seven gluteal activation exercises three times weekly prior to their normal training sessions. Whilst the CON group performed the conventional training with no gluteal activation exercises. Electromyography (EMG) was measured during a maximal isometric unilateral squat (MVIC) and unilateral hip extension force from the left and right vastus lateralis, gluteus maximus, and biceps femoris. After 6 weeks of training there was no significant main or interaction effect ($p > 0.05$) of EMG and peak force for MVIC and hip extension between GLUTE and CON. The current gluteal activation programme did not enhance EMG activity and hip extension force therefore, the body-weight exercises may not have been sufficient to elicit the appropriate changes.

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KEYWORDS
Gluteus maximus; muscle activation; electromyography; peak force

Introduction

The gluteus maximus (GM) is the largest of the gluteal muscles that produces powerful hip extension and externally rotates the hip (McAndrew, Gorelick, & Brown, 2006). It's large cross-sectional area provides the necessary force production for activities such as; running, sprinting, jumping, and side-stepping (Neumann, 2010). Efficient hip movement along with the recruitment of the GM and associated hip extensor muscles are required for optimal force production. Inefficient or incorrect movements can result in changes to muscle performance that can lead to developing impairments that may increase the risk of injury (Lewis, Sahrmann, & Moran, 2009) such as, anterior cruciate ligament sprains (Hewett, Myer, & Ford, 2006), iliotibial band friction syndrome (Fredericson et al., 2000), lower back pain (Hossain & Nokes, 2005), and patellafemoral pain (Cichanowski, Schmitt,

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Johnson, & Niemuth, 2007). In terms of muscle activation, recent research revealed that biceps femoris (BF) and GM had similar muscle activity levels and there was no dominant firing pattern when loading increased during deadlift resistance exercise (Cochrane & Barnes, 2015). Alternatively, previous studies have primarily focused on examining the activation level of various gluteal exercises to assist and identify the appropriate prescription for exercise treatment (Ayotte, Stetts, Keenan, & Greenqay, 2007; Boren et al., 2011; Distefano, Blackburn, Marshall, & Padua, 2009).

As a result of the purported link between hip dysfunction and lower limb injury, there has been increased inquiry on enhancing GM strength to be included in conditioning and rehabilitation programmes (Crow, Buttifant, Kearny, & Hrysomallis, 2012; Nakagawa, Moriya, Maciel, & Serrao, 2012; Wagner et al., 2010). The importance of gluteal activation allows clinicians to select exercises during the different stages of rehabilitation but knowledge about improving gluteal strength may be a greater determinant for sport performance, rehabilitation, and injury prevention. However, little is known about targeting the gluteal muscle group for strength gains to improve performance.

Presently, only two gluteal activation studies have been specifically designed to determine the use of gluteal exercises as a warm-up intervention to improve vertical jump performance (Crow et al., 2012; Healy & Harrison, 2014). Results have varied, using a seven exercise warm-up gluteal activation protocol Crow et al. (2012) reported a 4.2% increase in peak power vertical jump while the same warm-up gluteal protocol observed no significant changes in single-leg depth jump performance (Healy & Harrison, 2014). Currently, there is a paucity of information regarding the efficacy of gluteal training and to our knowledge no study has examined the efficacy of implementing a short-term gluteal activation programme to improve muscle force. The idea of strengthening the larger GM as the primary hip extensor has merit to reduce movement deficiency, musculoskeletal disorders, and possible injury while enhancing athletic performance. Therefore, the aim of this study was to investigate the effect of implementing a gluteal strengthening programme over a short term to determine whether neuromuscular changes in electromyography (EMG) was capable of improving muscle force during unilateral hip extension (UHE). Given the theoretical and anecdotal evidence of improving GM, we hypothesized that implementing a gluteal programme would enhance muscle performance.

Methods

Participants

Twenty four semi-professional rugby males volunteered for the study. In consultation with the strength and conditioning coach, the study was implemented in the middle of the club rugby season. In this period, the players were completing one game per week, two specific rugby trainings per week, four resistance trainings, and one speed/running session. During the experimental period (6 weeks), all players continued this conventional training programme. Players were assigned randomly to a gluteal activation group (GLUTE) (age 20.9 ± 3.1 year; height 184 ± 7.2 cm; body mass 100.2 ± 10.8 kg) or control group (CON) (age 22.2 ± 3.8 years; height 187 ± 8.1 cm; body mass 100.7 ± 13.9 kg). The exercises of the GLUTE group were primarily selected on the basis to target gluteal

group, specifically GM, although the side-to-side monster walks targeted the gluteus medius. Throughout the 6-week training intervention, the GLUTE and CON groups performed the same amount of training (eight sessions per week, equated in terms of intensity and volume); the only difference was that GLUTE group performed gluteal activation exercises three times weekly prior to their normal training sessions. The CON group performed the conventional training with no gluteal activation exercises. The strength and conditioning coach supervised the training and guaranteed that there was no significant difference in conventional training between the GLUTE and CON groups during the experimental period.

Players were excluded if they had a history of pain, injury, or surgery of the lumbar spine and hip over the last 6 months, any previous lower limb injuries, or orthopaedic contraindications to resistance training. Three out of the 24 participants indicated they were left leg dominate; determined by the leg they used to kick a rugby ball maximum distance. Ethical approval was granted by the University Human Ethics Committee, and written informed consent was obtained from players.

Study design

Players performed pre- and post-testing of a unilateral maximum isometric squat and unilateral closed-kinetic chain hip extension. Telemetric EMG measured muscle activity during the isometric unilateral squat and hip extension from the left and right vastus lateralis (VL), GM and BF. GLUTE involved intensive gluteal body-weight exercises that involved seven exercises taking approximately 10 min to complete that was performed 3x/week for 6 weeks. The seven exercises of the GLUTE programme (Figure 1) include hip airplane exercise that have been developed by McGill (2014) and have been further modified:

(1) Hip airplane with hip thrust, (2 sets, 20 reps for each leg)
(2) Hip leg airplane with trunk rotation, (2 sets, 20 reps for each leg)
(3) Hip leg airplane with a black Thera band resistance applied to the knee that included trunk rotation with DB held in the hand (2 sets, 10 reps for each leg)
(4) Monster walk (3 m) with black Thera band resistance secured around the forefoot (forward and back 3x) (Cambridge, Sidorkewicz, Ikeda, & McGill, 2012)
(5) Monster walk side to side (3 m) with black Thera band secured around the forefoot (3x) (Cambridge et al., 2012)
(6) Supine bridge with glute squeeze – lying on back, with flexed knees and toes touching the wall the GM is then contracted, the hips are then raised without the hamstrings contracting. Five reps of 5-s contractions of each leg (McGill, 2016)
(7) Single-leg stance with head nod – standing on one leg with eyes closed, gently nod the head up and down for 3 min; repeat on other leg.

For CON, no supplementary gluteal exercises were included with the player's current training. After 6 weeks, GLUTE and CON were retested on the same muscle strength tests. To negate the influence of prior activity on hip extension, no warm-up was performed and on arrival participants were instructed to passively rest in a chair for approximately 10 min. Further, players were instructed to avoid strenuous exercise, caffeine, and maintain

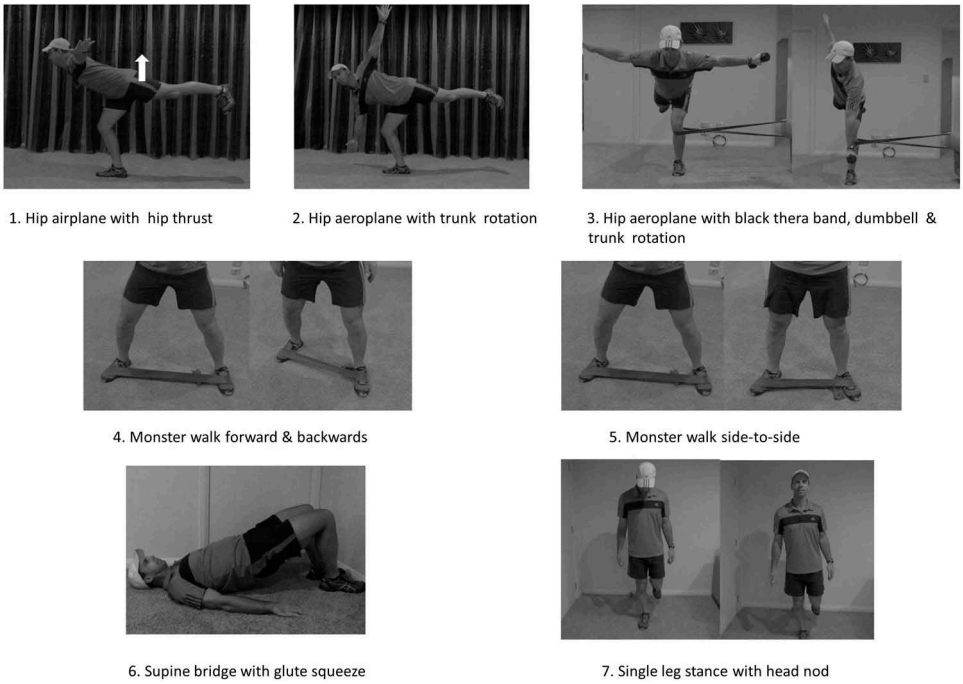

Figure 1. The Seven exercises of the Glute Programme.

their normal nutritional intake prior to testing. To account for daily biorhythms, all testing was conducted at the same time of day and participants were encouraged to maintain their dietary, sleeping, and drinking habits during the study.

Maximal voluntary isometric contraction

It is common practice to employ open-chain kinetic exercises such as the prone hip extension to assess maximal voluntary isometric contractions (MVIC) of the GM (Distefano et al., 2009; Ekstrom et al., 2007). However, the test lacks posture specificity, especially in sports such as rugby, which is predominately performed in an upright posture. To improve ecological validity, a closed-chain single-leg isometric squat activates the gluteus medius that assists frontal plane hip drive, which in turn, causes GM activation. Additionally, the core muscles that stabilize the lumbar spine are activated that provides maximum contraction of GM without causing increased lumbar spine movement (McGill, 2014). Thus, having the joints locked and loaded it was envisaged that the optimum force of GM can be exerted when the lumbar spine is stabilized and the force closure of the sacroiliac joint is maximized, which is difficult to achieve with traditional prone hip extension techniques. Therefore, we decided to base our MVIC on a closed-kinetic exercise related to rugby activity, where the hip was stabilized to assist optimal GM activation.

Using a hydraulic squat machine (AeroStrength Hydraulic Fitness Equipment, Durango, CO, U.S.A.), a portable force plate (Accugait, Advance Mechanical Technology Inc, MA, U.S.A.) was mounted to the squat machine platform, which measured the

ground reaction force of the MVIC. This was connected to a notebook computer (Toshiba, Tecra Z50-A, China) where the force data were recorded by a data acquisition system (NetForce version 2.4.0, Advance Mechanical Technology Inc, MA, U.S.A.) that sampled at 1000 Hz. peak force (N) was then analysed offline by BioAnalysis software (version 2.3.0 Advance Mechanical Technology Inc, MA, U.S.A.). The MVIC was performed by instructing players to position 1 foot in the centre of the force plate and place their shoulders under the padding where they assumed a hip and knee angle of 120°, which was measured by manual goniometer. They were informed to contract the GM while pushing their foot as forcefully as possible into the ground force plate. Two MVIC's of 4 s each were performed per leg that was separated by 30-s rest. An external trigger was used to synchronize the force plate and EMG receiver, which embedded an electronic marker into the data files that were subsequently stored offline on notebook computer and analysed using the BioAnalysis software.

Unilateral hip extension (UHE)

UHE was performed in a semi-orthostatic position on a hydraulic glute machine (AeroStrength Hydraulic Fitness Equipment, Durango, CO, U.S.A.) with the portable force plate being secured to the foot plate of the hip machine by a fastening belt. Players were instructed to place 1 foot in contact with the foot plate, at a hip and knee angle of 120°, respectively, that was measured by manual goniometer. The other leg was placed on the platform at a hip and knee angle of 120°. Every player positioned both forearms on the padded rest and grasped the support bars; with a flat back and the resistance of hydraulic cylinder set to level 5 (Cochrane & Harnett, 2015) players were instructed to maximally extend their hip as fast and forcefully as possible. Two repetitions of the right and left hip were measured and a 20-s passive rest separated each repetition with the hip being randomized. The highest (peak) value of left and right hip extension force of the concentric phase from the two repetitions and the associated EMG was used for subsequent analysis.

EMG

Muscle activity of the left and right VL, BF, and GM muscles were measured using surface EMG. Prior to electrode placement, the area was shaved, gently abraded and cleaned with isopropyl alcohol. Surface pre-gelled Ag–AgCl electrodes, 30 × 22-mm diameter (Ambu, Ballerup, Denmark) were placed over the mid-belly of the muscle at an inter-electrode distance of 20 mm parallel to the direction of the fibres according to Surface EMG for the Non-Invasive Assessment of Muscles recommendations (Hermens, Freriks, Disselhorst-Klug, & Rau, 2000). The electrodes were connected to wireless EMG sensors that were securely fastened to the muscle sites with adhesive tape. The sensors pre-amplified the EMG signals using a first-order high-pass filter (10 Hz) at a gain of 400. The EMG signals were then transmitted telemetrically in real time to a PC interface receiver (Telemyo DTS, Noraxon, Scottsdale, AZ, U.S.A.) and were recorded by a data acquisition system (MyoResearch XP Master, version 1.07.1, Noraxon, Scottsdale, AZ, U.S.A.). Data were filtered with a low-pass filter of 500 Hz and sampled at 3000 Hz, which was synchronized with video capture (Logitech C920 HD Pro Webcam, NSW, Australia).

Table 1. Mean ± SD of absolute EMG_{rms} VL, BF, GM of left and right UHE for Glute and Control.

		Left UHE			Right UHE		
		VL (μV)	BF (μV)	GM (μV)	VL (μV)	BF (μV)	GM (μV)
Glute	Pre	77.5 ± 27.3	116.2 ± 28.6	56.6 ± 28.3	55.9 ± 27.3	142.5 ± 25.3	70.4 ± 23.1
	Post	105.0 ± 31.0	162.6 ± 26.7	95.8 ± 23.0	120.8 ± 26.7	147.5 ± 30.8	96.6 ± 26.4
Control	Pre	116.7 ± 35.5	124.0 ± 34.4	88.5 ± 26.3	103.2 ± 26.1	99.4 ± 23.0	97.7 ± 20.3
	Post	141.7 ± 33.7	163.3 ± 34.1	92.6 ± 29.9	116.9 ± 31.7	120.9 ± 29.9	132.2 ± 24.5

Table 2. Mean ± SD of normalized EMG_{rms} VL, BF, GM of left and right UHE for Glute and Control.

		Left UHE			Right UHE		
		VL (%)	BF (%)	GM (%)	VL (%)	BF (%)	GM (%)
Glute	Pre	51.9 ± 24.7	70.5 ± 42.1	70.8 ± 17.2	42.2 ± 29.2	88.2 ± 55.4	82.7 ± 57.3
	Post	57.9 ± 15.8	77.0 ± 64.6	63.9 ± 30.9	52.1 ± 20.3	93.9 ± 76.1	83.1 ± 45.3
Control	Pre	77.3 ± 56.4	78.8 ± 49.6	71.5 ± 45.8	50.4 ± 23.7	82.8 ± 29.6	57.9 ± 18.2
	Post	69.0 ± 29.6	85.2 ± 12.0	79.7 ± 24.2	77.8 ± 44.7	84.2 ± 18.4	70.7 ± 13.3

Subsequently, the absolute EMG data were rectified and smoothed over 100 ms (root means square algorithm, Table 1) and were normalized to the MVIC of VL, BF, and GM. The EMG_{rms} mean amplitude of VL, BF, and GM (Table 2) were analysed by MyoResearch software (XP Master, version 1.07.1, Noraxon, Scottsdale, AZ, U.S.A.,) during the concentric phase of hip extension. This was determined from the initial hip position (120° hip flexion) to the individual's maximum hip extension.

Statistical analyses

A three-way repeated measure analysis of variance (ANOVA) was used to evaluate group (GLUTE vs. CON) and hip (left vs. right) over time (pre vs. post) for MVIC and UHE. For EMG_{rms}, a repeated ANOVA assessed changes over time in group, hip, and muscle sites (VL, GM, BF). If a significant main effect of group, hip, time, or an interaction was observed, post hoc pairwise comparisons were performed using the Bonferroni adjustment. A two-tailed, paired Students *T*-test was performed to assess age, height, and weight between GLUTE and CON groups. All statistical analysis was computed by using SPSS for Windows (version 20.0, IBM, New York, U.S.A.). The level of significance was set at $p \leq 0.05$ and the values are reported as mean ± standard deviation.

Results

At the commencement of the study, there was no statistical difference ($p > 0.05$) in height, weight, and age between GLUTE and CON. Further, there was no initial difference in MVIC and UHE peak force of the right and left hip between GLUTE and CON. Post-training (Figures 2 and 3) revealed that there was no significant main or interaction effect ($p > 0.05$) of peak force and EMG_{rms} for MVIC and UHE between GLUTE and CON (Table1).

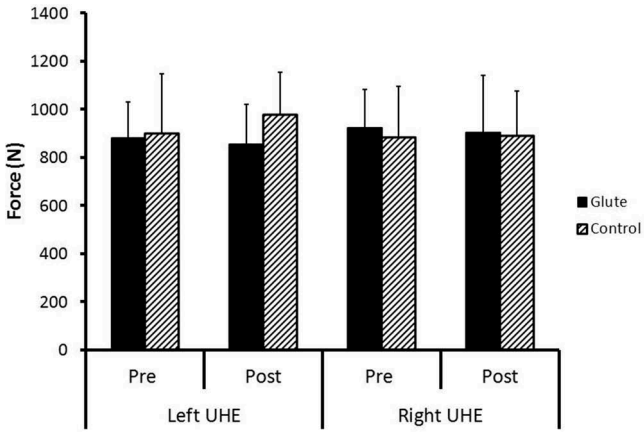

Figure 2. Left and right UHE muscle force for Glute and Control.

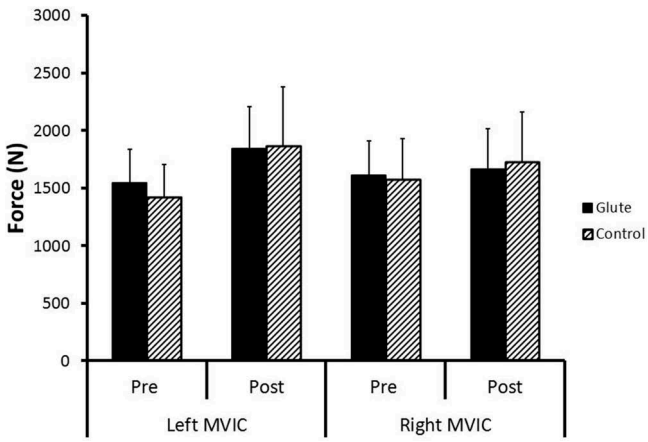

Figure 3. Left and right MVIC force for Glute and Control.

Discussion

The aim of the study was to determine if short-duration gluteal activation programme was able to facilitate neuromuscular activation to enhance muscular force. The results revealed that 18 sessions (3x/week for 6 weeks) that targeted gluteal activation, in addition to the players' conventional training, was unable to enhance UHE force.

An increase in strength during the early stages of resistance training is largely attributed to neural changes (Gabriel, Kamen, & Frost, 2006; Sale, 1988) where EMG amplitude has been reported to increase during resistance training with a parallel enhancement in muscle performance (Aagaard, Simonsen, Andersen, Magnusson, & Dyhre-Poulsen, 2002; Hakkinen et al., 1998). However, in the current study there was no apparent change in EMG activity and muscle force after 6 weeks of gluteal training. The capacity to further increase neural drive may have been reduced by the training

status of the participants, who were well-trained players; with their developed neuromuscular system it may have prevented any additional improvement.

Contrary to the current findings, the implementation of an acute gluteal activation protocol reported a significant increase in countermovement vertical jump peak power compared with a whole-body vibration and dynamic warm-up in elite Australian Rules football players (Crow et al., 2012). The authors claim that the findings were directly related to the gluteal exercises that enhanced muscle activation. However, using the same seven gluteal exercises as Crow et al. (2012), Healy and Harrison (2014) found that in well-trained track athletes that performed 10 repetitions of single-leg depth jumping failed to acutely enhance muscle performance and biomechanical jump measures (Healy & Harrison, 2014). The discrepancy between the aforementioned acute studies and the current study can be explained by the experimental design of acute versus short-term training. Additionally, the seven gluteal exercises utilized by Crow et al. (2012) and Healy and Harrison (2014) differ to the current gluteal protocol that included dynamic and isometric exercises. The current gluteal programme was carefully selected to specifically target gluteal activation but other biomechanically similar exercises of UHE and its dynamic nature could have been considered.

GM is a powerful hip extensor and also acts to external rotator of the hip (McAndrew et al., 2006) that can be divided into three distinctive areas – the upper third of the muscle comprises predominately of slow twitch fibres that assists hip abduction (McAndrew et al., 2006); the middle and lower regions are thought to have a higher proportion of fast twitch fibres giving them importance for rapid hip extension, external rotation (McAndrew et al., 2006) and assists hip adduction (Floyd, 2009). The current surface EMG was positioned at 50% on a line between the sacral vertebrae and the greater trochanter that corresponded with the greatest prominence of the middle gluteal region; however, in the future it would be of interest to analyse EMG activity of the upper GM region to determine if gluteal exercises are able to influence this area.

When comparing participants and muscles across various trials, it is common practice to normalize EMG (Worrell et al., 2001). In accordance to normalized procedures, a referenced MVIC is selected and the joint angle of the normalized contraction should be similar to the activity of interest (Worrell et al., 2001). In the current study, the knee and hip angle (120°) was identical between MVIC and UHE further, to improve ecological validity we decided to use a closed-kinetic MVIC exercise that was applicable to rugby. However, there was some variability during the MVIC, where certain players may have favoured more BF or GM recruitment, which may have influenced the EMG recording of UHE.

In a semi-professional team environment, there are time constraints of including extra training; thus body-weight exercises were chosen for its ease to be included into the player's current training. Although this is an important practical consideration, body-weight exercises may not be sufficient to elicit the appropriate changes over a 6-week period; often it is difficult to increase the load of body-weight exercises without external assistance; therefore, having a greater and progressive loading regime may have assisted improvement. Moreover, the intensity of the gluteal exercises of the prescribed repetitions, sets, and frequency may have not been appropriate to elicit the appropriate changes during 6 weeks of training.

Conclusion

Six weeks of gluteal activation programme performed in addition to the players normal training did not enhance EMG activity and UHE force. However, the rationale of the topic requires further investigation to determine if short-term and long-term gluteal activation is viable option for conditioning and rehabilitation purposes.

Disclosure statement

No potential conflict of interest was reported by the authors.

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